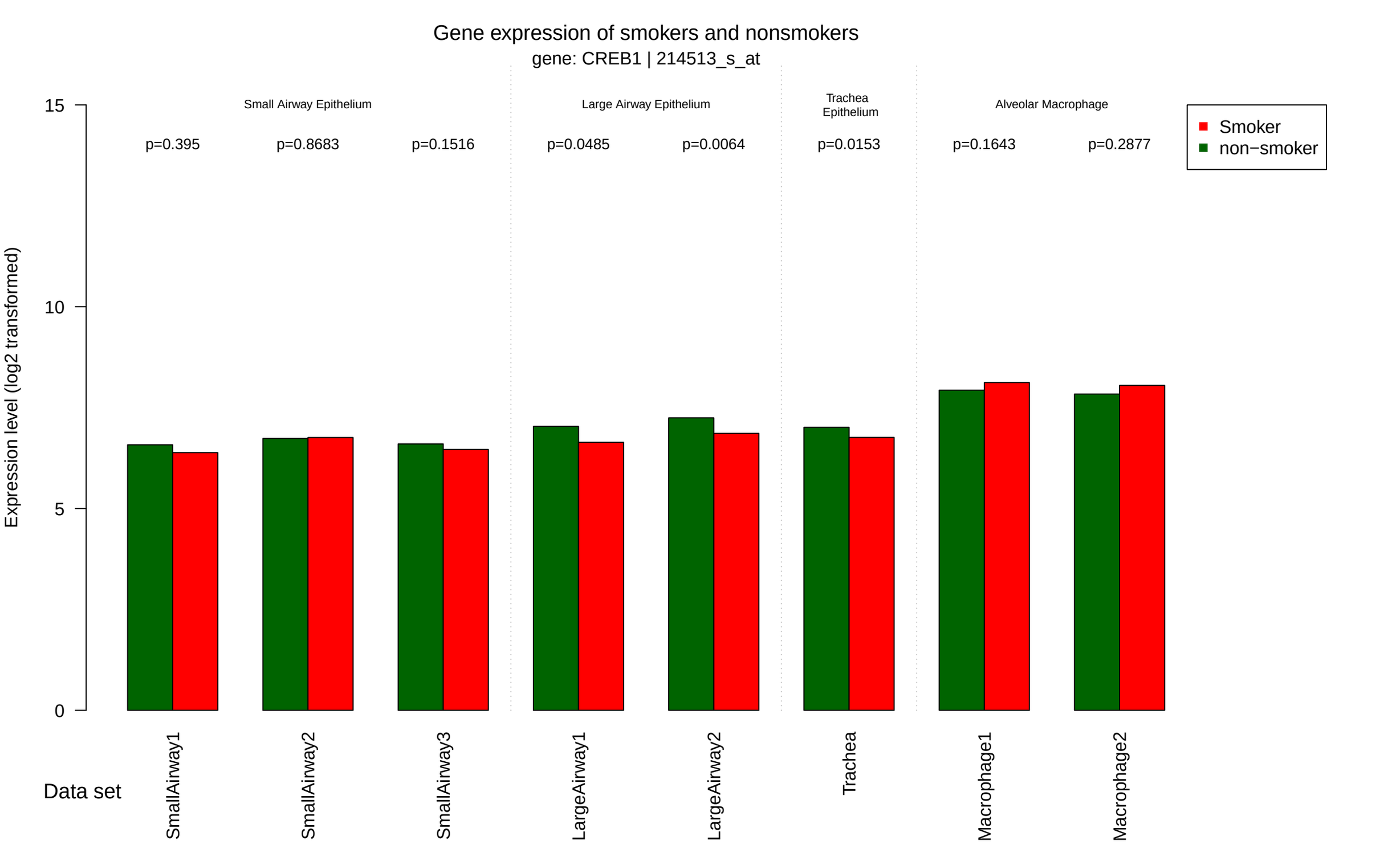
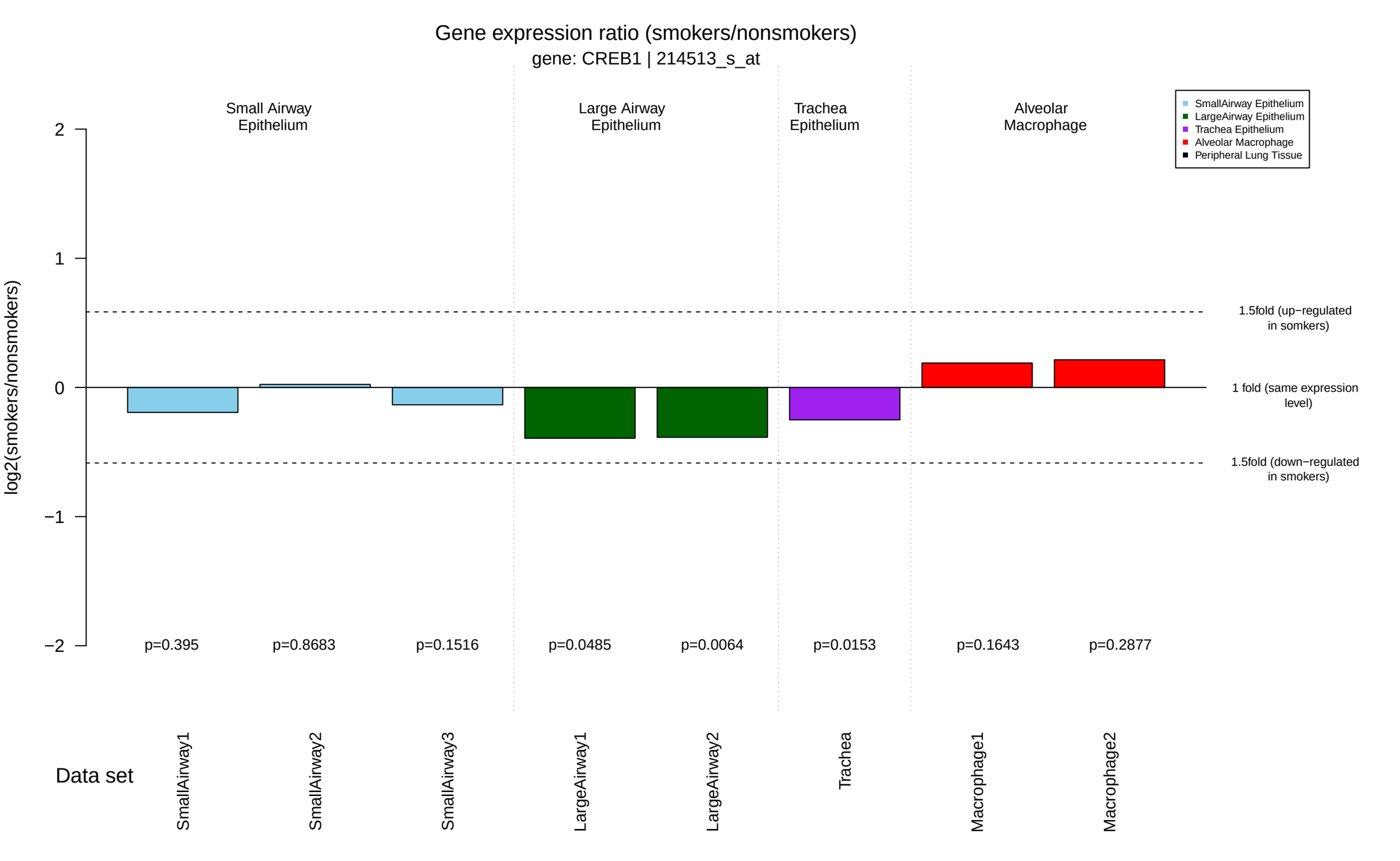
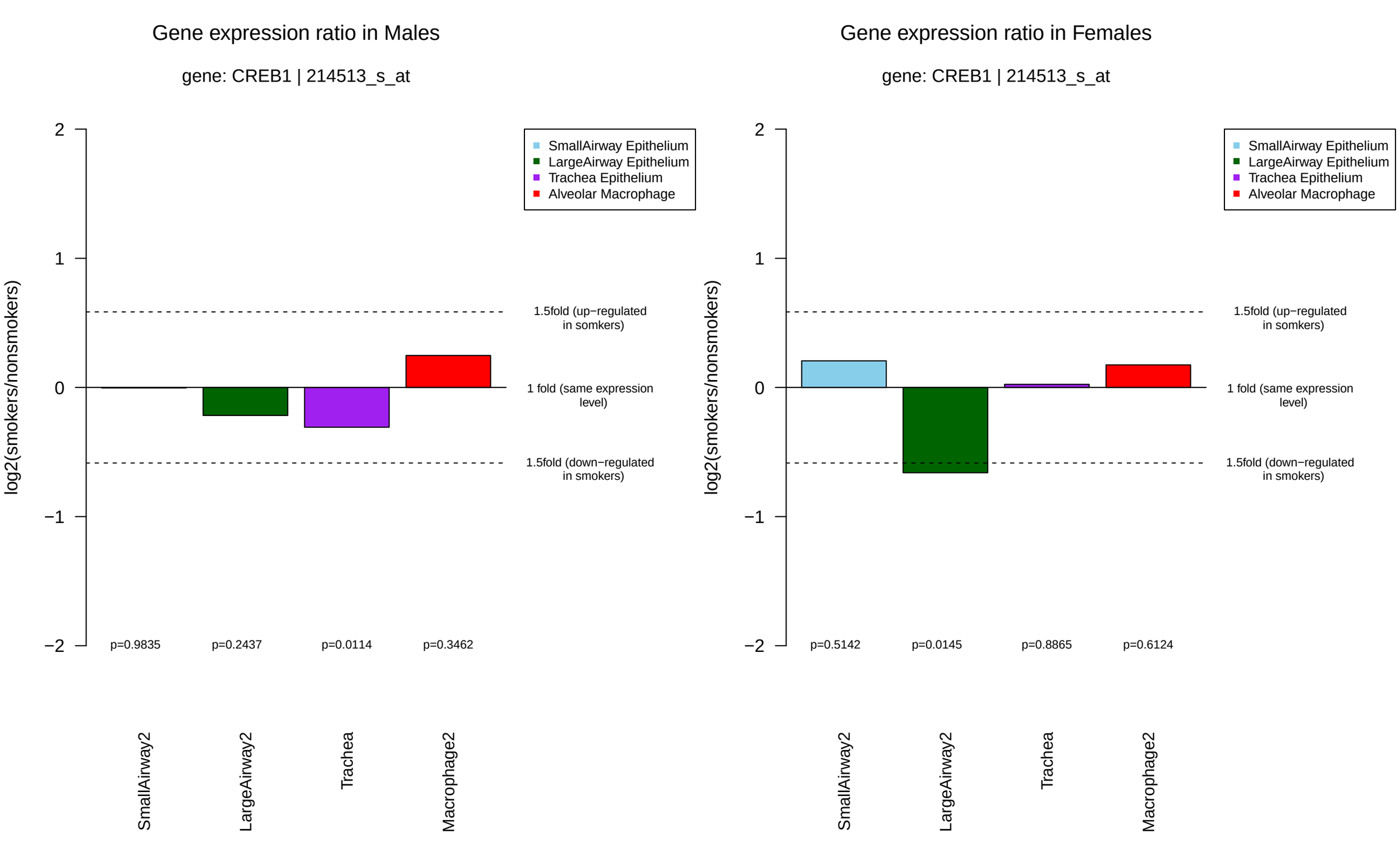
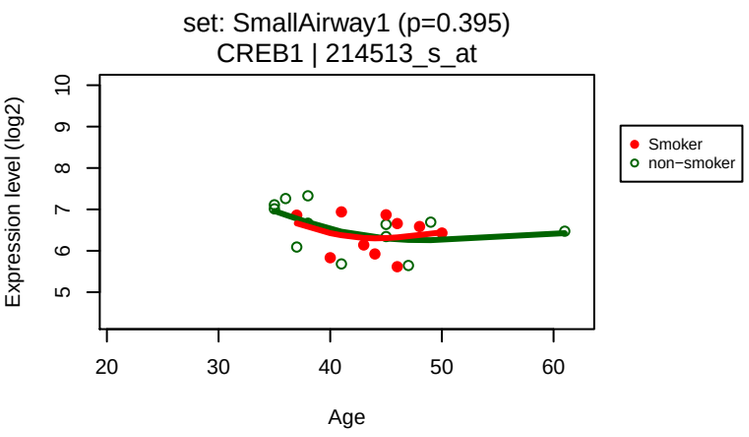
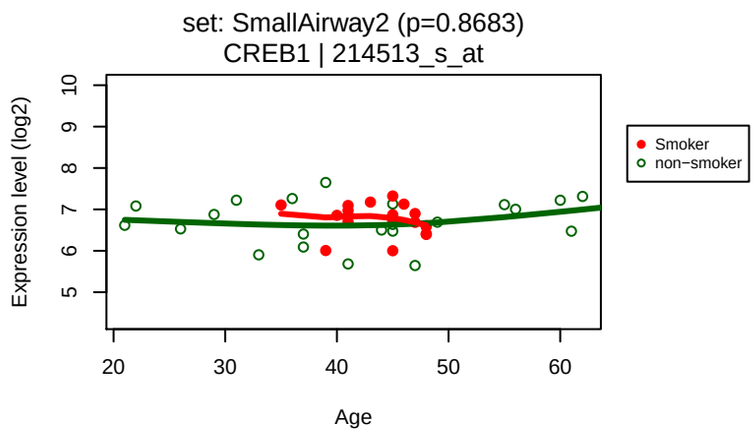
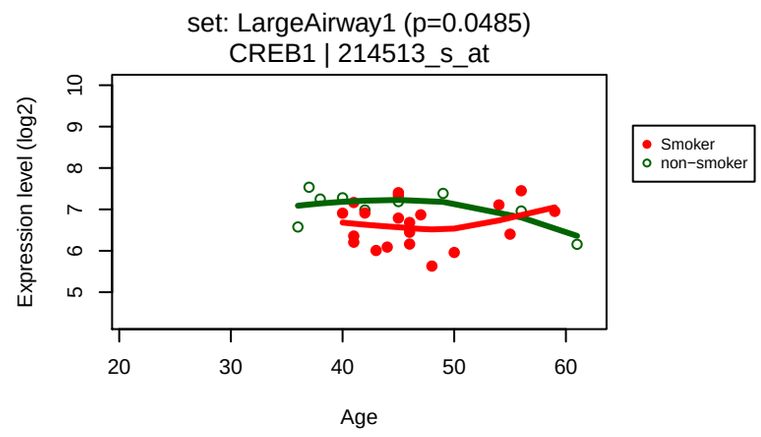

Gene expression of smokers and nonsmokers
gene: CREB1 | 214513_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.395
p=0.8683
p=0.1516
p=0.0485
p=0.0064
p=0.0153
p=0.1643
p=0.2877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

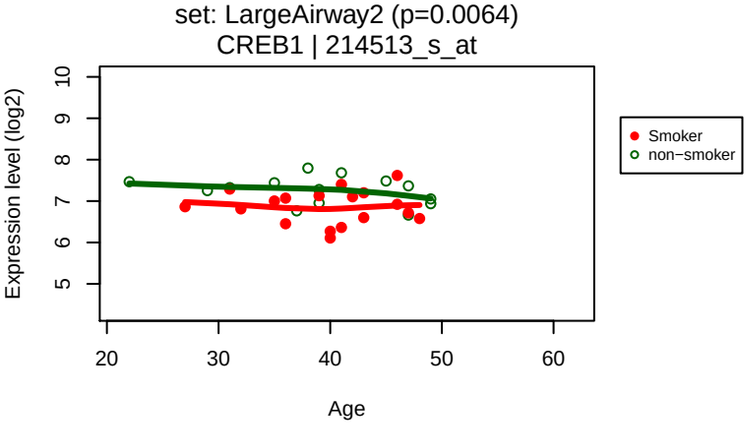

Gene expression ratio (smokers/nonsmokers)
gene: CREB1 | 214513_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.395
p=0.8683
p=0.1516
p=0.0485
p=0.0064
p=0.0153
p=0.1643
p=0.2877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

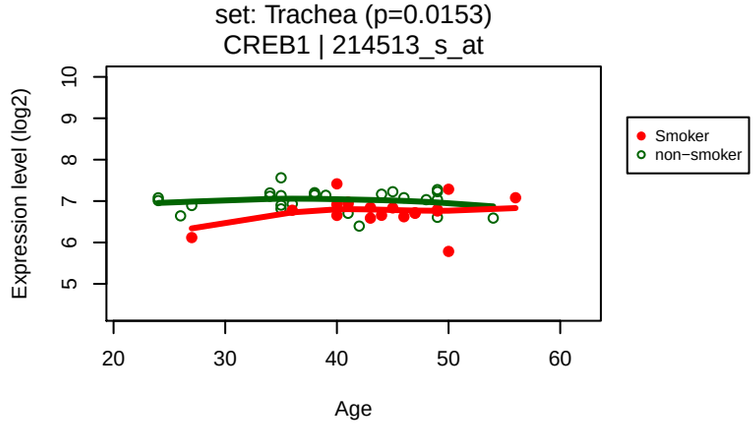

Gene expression ratio in Males
gene: CREB1 | 214513_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9835
p=0.2437
p=0.0114
p=0.3462
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB1 | 214513_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5142
p=0.0145
p=0.8865
p=0.6124
SmallAirway2
LargeAirway2
Trachea
Macrophage2

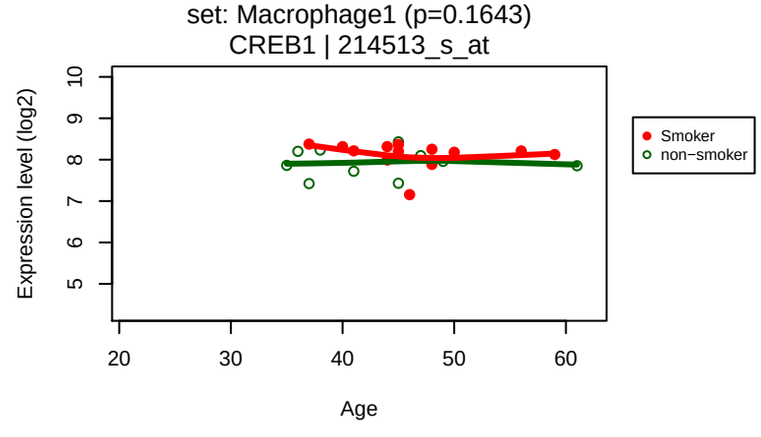

set: SmallAirway1 (p=0.395)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker

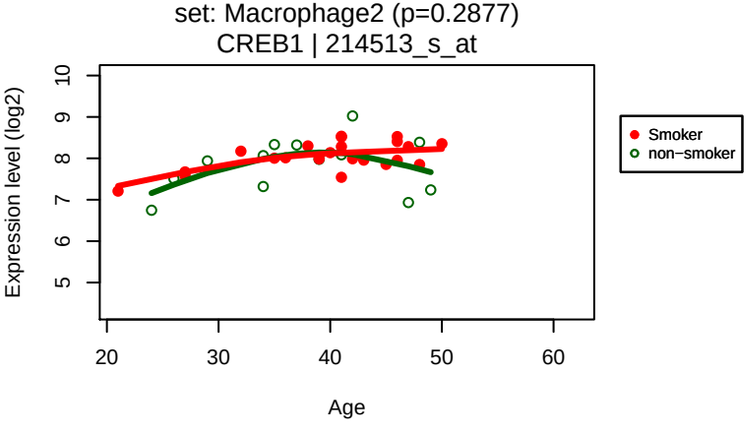

set: SmallAirway2 (p=0.8683)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker

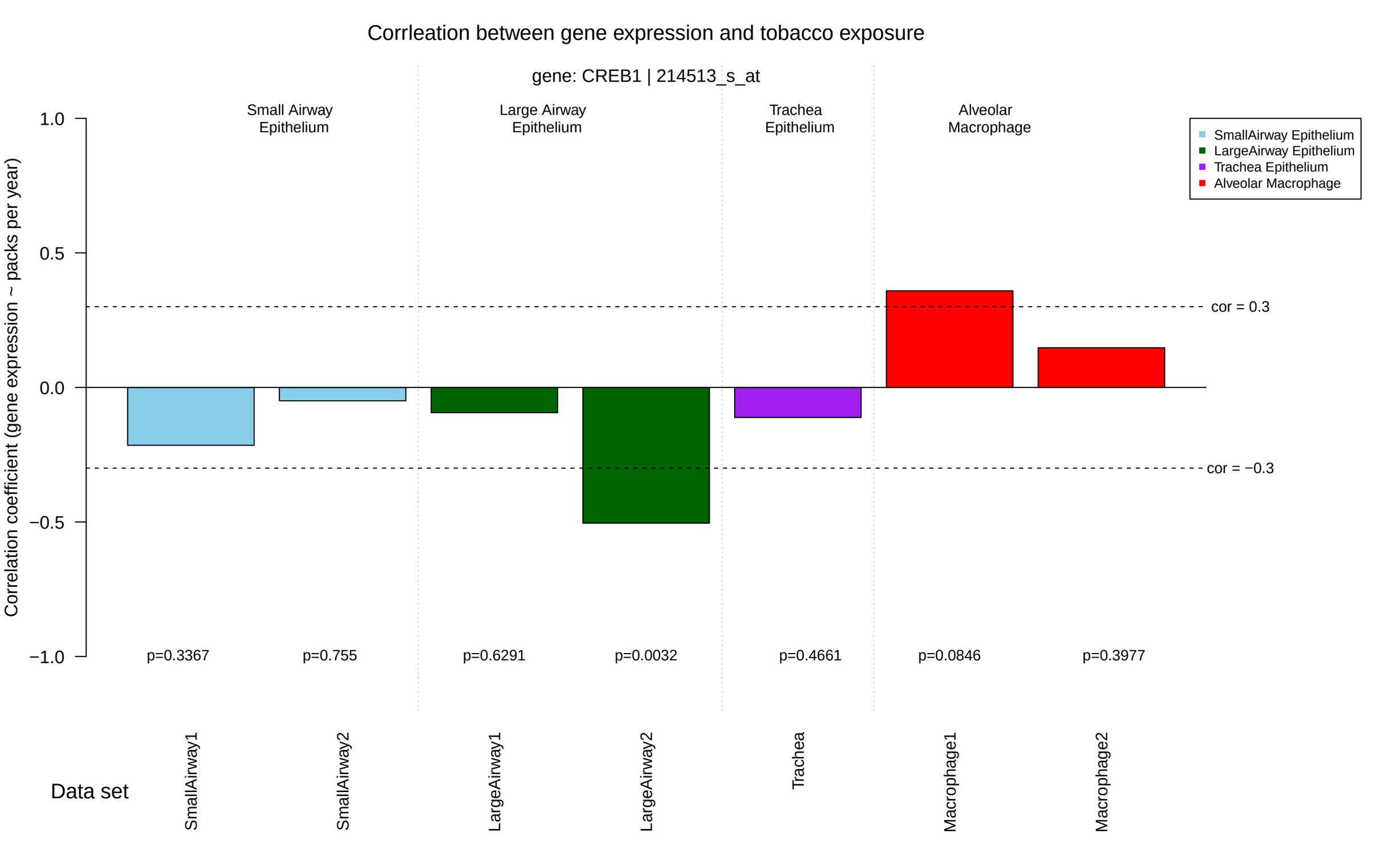

set: LargeAirway1 (p=0.0485)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0064)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0153)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1643)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2877)
CREB1 | 214513_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB1 | 214513_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3367
p=0.755
p=0.6291
p=0.0032
p=0.4661
p=0.0846
p=0.3977
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2